3.7 $\pi^2 \leq 10$

WORKSHEET

The question

STEP II 2014, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q2

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why might a function that is zero at both ends of $[0, \pi]$ have to be steep if it is large in the middle? Why does the condition $f(0) = f(\pi) = 0$ matter?

STAGE 1 Sine waves

Ask yourself What is $\int_0^\pi \sin^2 nx \, dx$? And $\int_0^\pi \cos^2 nx \, dx$? (Use $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$.)

STAGE 2 The end conditions matter

Ask yourself Find a simple function with $f(0) \neq 0$ for which $(*)$ fails. Then find one with $f(0) = 0$ but $f(\pi) \neq 0$.

STAGE 3 A parabola

Ask yourself Which quadratic vanishes at $x = 0$ and $x = \pi$? (Up to a constant multiple, there is only one. Why doesn't the multiple matter?)

STAGE 4 A trigonometric choice

Ask yourself For $f(x) = p \sin \frac{1}{2}x + q \cos \frac{1}{2}x + r$, what conditions do $f(0) = 0$ and $f(\pi) = 0$ impose on p, q, r?

Pause and try Compute $\int_0^\pi f^2$ and $\int_0^\pi (f')^2$ for this function. What inequality for π results?

What's really going on

A proof with Fourier series

Why is $(*)$ true? At university you will learn that any reasonable function on $[0, \pi]$ that vanishes at both ends can be written as a *Fourier sine series*,

$$\mathrm{f}(x) = b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \cdots .$$

Because $\int_0^\pi \sin mx \sin nx \, \mathrm{d}x = 0$ for $m \neq n$, all the cross terms vanish when you square and integrate, giving

$$\int_0^\pi \mathrm{f}^2 = \frac{\pi}{2}(b_1^2 + b_2^2 + b_3^2 + \cdots), \qquad \int_0^\pi (\mathrm{f}')^2 = \frac{\pi}{2}(1^2 b_1^2 + 2^2 b_2^2 + 3^2 b_3^2 + \cdots).$$

Each term on the right is at least the corresponding term on the left, which proves $(*)$, and shows that equality holds only when f is a multiple of $\sin x$. Part (i) of the question is the single-term case.

Vibrating strings and quantum particles

The ratio $\int (\mathrm{f}')^2 / \int \mathrm{f}^2$ is called a *Rayleigh quotient*, and $(*)$ says its smallest possible value, for functions pinned at 0 and π, is 1. This number has physical meaning. For a string fixed at both ends, $\int (\mathrm{f}')^2$ measures the stretching energy and $\int \mathrm{f}^2$ the size of the displacement, and the minimum value of the ratio determines the lowest note the string can play. In quantum mechanics, the same minimum is the lowest energy of a particle trapped in a box: it cannot be zero, because a wave pinned at both ends must bend. Lord Rayleigh used exactly this trick, guessing a test function and computing the ratio, to estimate the frequencies of vibrating systems; because the extreme case gives the minimum, any guess gives an upper bound on the true lowest frequency, and good guesses give very accurate ones.

The calculus of variations

Finding the function that minimises a ratio of integrals is a problem in the *calculus of variations*, where the unknown is a whole function rather than a number. The minimiser here, $\sin x$, satisfies the differential equation $\mathrm{f}'' + \mathrm{f} = 0$, and this is no coincidence: minimising an integral always leads to a differential equation for the minimiser. We return to this idea in Section 4.7.

Try this next

1. **A quartic.** Use $\mathrm{f}(x) = x^2(\pi - x)^2$ in $(*)$. Show that it gives only $\pi^2 \leq 12$. Why is this test function worse than the parabola? (Compare its shape with $\sin x$.)

2. **Two sine waves.** Check $(*)$ directly for $\mathrm{f}(x) = \sin x + \sin 2x$, computing both sides.

3 A different interval. Show by a substitution that if $g(0) = g(L) = 0$ then $\displaystyle\int_0^L g^2 \le \frac{L^2}{\pi^2}\int_0^L (g')^2$. What does this say about strings of different lengths?

4 Rayleigh's method. HARDER Take $f(x) = x(\pi - x) + \lambda x^2(\pi - x)^2$ and choose λ to make the bound on π^2 as good as possible. (The algebra is heavy; a computer algebra system helps.)

Further reading

- **T. W. Körner, *Fourier Analysis*** (Cambridge University Press). A wonderful, discursive book on Fourier series and their applications, written by a Cambridge analyst.
- **Hardy, Littlewood & Pólya, *Inequalities*** (Cambridge University Press). Chapter 7 discusses Wirtinger's inequality and its relatives.
- **Richard Feynman, *The Feynman Lectures on Physics*,** Volume II, Chapter 19 ("The principle of least action"). Feynman's lecture on minimising integrals, with physical intuition throughout. Free online.

The full guided solution is in Section 3.7:
https://beyondstep.keithmcnulty.org/ch03/wirtinger.html